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Code-C Phase-1

Aakash

Medical|IIT-JEE|Foundations

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MM : 240

TYM for NEET-2025-26_Practice Test-01C

Time : 60 Min.

PHYSICS

- | | |
|--------|---------|
| 1. (3) | 9. (2) |
| 2. (1) | 10. (2) |
| 3. (2) | 11. (2) |
| 4. (2) | 12. (3) |
| 5. (3) | 13. (2) |
| 6. (1) | 14. (2) |
| 7. (3) | 15. (3) |
| 8. (3) | |

CHEMISTRY

- | | |
|---------|---------|
| 16. (4) | 24. (3) |
| 17. (3) | 25. (4) |
| 18. (2) | 26. (4) |
| 19. (1) | 27. (4) |
| 20. (1) | 28. (3) |
| 21. (1) | 29. (4) |
| 22. (3) | 30. (1) |
| 23. (2) | |

BOTANY

- | | |
|---------|---------|
| 31. (1) | 39. (2) |
| 32. (2) | 40. (3) |
| 33. (3) | 41. (3) |
| 34. (1) | 42. (3) |
| 35. (2) | 43. (2) |
| 36. (4) | 44. (1) |

37. (1)

45. (3)

38. (1)

ZOOLOGY

46. (4)

54. (3)

47. (2)

55. (4)

48. (2)

56. (3)

49. (2)

57. (1)

50. (3)

58. (2)

51. (4)

59. (1)

52. (2)

60. (4)

53. (4)



Hints and Solutions

PHYSICS

(1) Answer : (3)

Solution:

Least count = 1 MSD – 1 VSD

Given 9 MSD = 10 VSD

$$1 \text{ VSD} = \frac{9}{10} \text{ MSD}$$

$$\text{L.C.} = 1 \text{ MSD} - \frac{9}{10} \text{ MSD}$$

$$= \frac{1}{10} \text{ MSD}$$

$$= \frac{a}{10}$$

(2) Answer : (1)

Solution:

$$\text{Velocity} = [M^0 L T^{-1}]$$

$$x = 0, y = 1, z = -1$$

$$x + y + z = 0$$

(3) Answer : (2)

Hint:

$$\text{For addition } \Delta C_{eq} = \Delta C_1 + \Delta C_2$$

Solution:

$$C_{eq} = [(C_1 + C_2) \pm (\Delta C_1 + \Delta C_2)] \mu F$$

$$= [(10 + 20) \pm (1 + 1)] \mu F$$

$$= [30 \pm 2] \mu F$$

(4) Answer : (2)

Solution:

$$\text{Power} = \frac{\text{Work}}{\text{Time}} = [ML^2 T^{-3}]$$

$$\text{Energy} = [ML^2 T^{-2}]$$

$$\text{Force} = [ML T^{-2}]$$

$$\text{Momentum} = [ML T^{-1}]$$

(5) Answer : (3)

Solution:

$$\rho = \frac{m}{V} = \frac{m}{L^3}$$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + \frac{3\Delta L}{L}$$

$$= 3\% + 3 \times 2\% = 9\%$$

(6) Answer : (1)

Solution:

$$Z = (AB)^{\frac{1}{2}}$$

$$Z = (2 \times 2)^{\frac{1}{2}} = 2$$

$$\Delta Z = \frac{1}{2} \left(\frac{0.1}{2} + \frac{0.1}{2} \right) \times 2$$

$$\Delta Z = 0.1$$

$$(AB)^{\frac{1}{2}} = 2.0 \text{ m} \pm 0.1 \text{ m}$$

(7) Answer : (3)

Solution:

$$[\alpha T^{3/2}] = [L T^{-1}]$$

$$[\alpha] = [L T^{-1-3/2}] = [L T^{-5/2}]$$

(8) Answer : (3)

Solution:

$$n_1 [M_1 L_1 T_1^{-2}] = n_2 [M_2 L_2 T_2^{-2}]$$

$$n_2 = n_1 \left[\frac{M}{2M} \times \frac{L}{2L} \times \left(\frac{2T}{T} \right)^2 \right]$$

$$n_2 = n_1$$

(9) Answer : (2)

Solution:

In a number with decimal, the trailing zero(s) are significant.

(10) Answer : (2)

Hint:

$$180^\circ = \pi \text{ radian}$$

Solution:

$$1^\circ = \frac{\pi}{180} \text{ radian} \Rightarrow 30^\circ = \frac{\pi}{6} \text{ radian}$$

(11) Answer : (2)

Solution:

Smaller is the least count, greater is the precision.

Reading 20.01 m is precise upto 0.01 m.

(12) Answer : (3)

Solution:

$$100 + 0.423 = 100.423$$

≈ 100 (upto correct decimal place)

(13) Answer : (2)

Hint:

$$Z = a + b, \Delta Z = \Delta a + \Delta b$$

Solution:

$$Z = P + Q$$

$$\Delta Z = \Delta P + \Delta Q$$

$$\% \text{ error} = \frac{\Delta Z}{Z} \times 100$$

$$= \frac{\Delta P + \Delta Q}{P + Q} \times 100$$

$$= \left(\frac{\Delta P}{P + Q} + \frac{\Delta Q}{P + Q} \right) \times 100$$

(14) Answer : (2)

Solution:

$$P = Fv$$

$$= [MLT^{-2}] [LT^{-1}]$$

$$\Rightarrow P = [ML^2T^{-3}]$$

$$1000 = M(10)^4 (1)^{-3}$$

$$M = 10^{-1} \text{ kg}$$

(15) Answer : (3)

Solution:

Random error can be minimised by taking large number of readings.

CHEMISTRY

(16) Answer : (4)

Solution:

Pure substance have a fixed composition. Example: Carbon dioxide

(17) Answer : (3)

Solution:

In nuclear reactions, mass is converted into energy.

(18) Answer : (2)

Solution:

$$^\circ\text{C} = \frac{5}{9} (^\circ\text{F} - 32) = -17.77$$

(19) Answer : (1)**Solution:**

$$1 \text{ pico} = 10^{-12}$$

(20) Answer : (1)**Solution:**

$$232.508 = 2.32508 \times 10^2$$

(21) Answer : (1)**Solution:**

Zeros preceding to first non-zero digit are not significant. 0.0033 has 2 significant figures.

(22) Answer : (3)**Solution:**

C	52.94	$\frac{52.94}{12} = 4.41$	$\frac{4.41}{2.94} = 1.5$	3
O	47.06	$\frac{47.06}{16} = 2.94$	$\frac{2.94}{2.94} = 1$	2

Empirical formula : C_3O_2 **(23) Answer : (2)****Hint:**

Mass percentage of oxygen

$$= \frac{\text{Weight of oxygen}}{\text{Weight of Compound}} \times 100$$

Solution:

Mass percentage of oxygen

$$= \frac{\text{Weight of oxygen}}{\text{Weight of compound}} \times 100$$

$$= \frac{64}{98} \times 100 = 65.31\%$$

(24) Answer : (3)**Hint:**Number of atoms = Number of moles $\times$ Atomicity $\times N_A$ **Solution:**Number of moles of $\text{S}_8 = 0.5$ mole

$$\text{Number of S atoms} = 0.5 \times 8 \times 6.022 \times 10^{23}$$

$$= 2.41 \times 10^{24} \text{ atoms}$$

(25) Answer : (4)**Solution:**

$$n_{\text{H}_2\text{O}} = \frac{36}{18} = 2$$

$$n_{\text{CH}_4} = \frac{64}{16} = 4$$

$$n_{\text{CO}_2} = \frac{44}{44} = 1 \text{ mole}$$

$$n_{\text{H}_2} = \frac{10}{2} = 5 \text{ moles}$$

(26) Answer : (4)**Hint:**Number of atoms = moles $\times$ atomicity $\times N_A$ **Solution:**

$$(1) \text{ N}_2 : \frac{14}{28} \times 2 \times N_A = N_A$$

$$(2) \text{ SO}_2 : \frac{64}{64} \times 3 \times N_A = 3N_A$$

$$(3) \text{ CO}_2 : \frac{88}{44} \times 3 \times N_A = 6N_A$$

$$(4) \text{ H}_2 : \frac{8}{2} \times 2 \times N_A = 8N_A$$

(27) Answer : (4)**Hint:**

$$\text{Mole of molecule/atom} = \frac{\text{Mass of molecule / atom}}{\text{Molar mass of molecule / atom}}$$

Solution:

$$\text{Mole of He} = \frac{2x}{4}$$

$$\text{Mole of N}_2 = \frac{x}{28}$$

$$\frac{\text{Mole of He}}{\text{Mole of N}_2} = \frac{\frac{2x}{4}}{\frac{x}{28}} = \frac{2x}{4} \times \frac{28}{x} = 14 : 1$$

(28) Answer : (3)**Solution:**

SI unit of thermodynamic temperature and Luminous intensity are Kelvin and Candela respectively.

(29) Answer : (4)**Solution:**

Compound	Molecular formula	Empirical formula
CH ₃ COOH	C ₂ H ₄ O ₂	CH ₂ O
HOCH ₂ CH ₂ OH	C ₂ H ₆ O ₂	CH ₃ O
C ₆ H ₆	C ₆ H ₆	CH
CH ₃ CHO	C ₂ H ₄ O	C ₂ H ₄ O

(30) Answer : (1)**Solution:**

$$1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$$

BOTANY

(31) Answer : (1)**Solution:**

Glycocalyx as a loose sheath is called slime layer and glycocalyx as a thick and tough covering is called capsule.

(32) Answer : (2)**Solution:**

Bacteria reproduce mainly by fission.

(33) Answer : (3)**Solution:**

Glycogen granule is inclusion body found in prokaryotes. Polysome has several ribosomes on mRNA.

(34) Answer : (1)**Solution:**

Several ribosomes may attach to a single mRNA and form a chain called polysome. The ribosomes of a polysome translate the mRNA into proteins.

(35) Answer : (2)**Solution:**

Amount of proteins are more than the amount of lipids in the membrane of erythrocytes.

(36) Answer : (4)**Solution:**

The fluid nature of membrane is important from the point of view of functions like cell growth, formation of intercellular junction, secretion, endocytosis, cell division, etc.

Cell wall protects the cell from attack of pathogens.

(37) Answer : (1)**Solution:**

In addition to phospholipids, cell membrane also contains proteins, carbohydrates and cholesterol. Cellulose is a component of cell wall.

(38) Answer : (1)**Solution:**

Hint: Many bacteria have circular DNA outside the genomic DNA. These DNA are called plasmid.

Sol.: Plasmid DNA is smaller than genomic DNA.

(39) Answer : (2)

Solution:

A bacterial flagellum is composed of filament, hook and basal body

(40) Answer : (3)

Solution:

Mesosomes are involved in respiration in prokaryotes.

(41) Answer : (3)

Solution:

The cell wall in a bacterium provides a strong structural support and prevent them from bursting or collapsing.

(42) Answer : (3)

Hint:

It is a non-membrane-bound cell organelle.

Solution:

Ribosome is common in both animal and plant cells.

(43) Answer : (2)

Solution:

Ribosomes are the sites of protein synthesis.

(44) Answer : (1)

Hint:

Slime layer protects the cell from loss of water and nutrients.

Solution:

Cell wall prevents the bacterium from bursting or collapsing. But capsule helps bacterium to hide from host's immune system.

(45) Answer : (3)

Solution:

Rudolf Virchow explained that cells divide and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*).

The concept of *Omnis cellula-e cellula* given by Rudolf Virchow suggests that all cells arise from pre-existing cells.

ZOOLOGY

(46) Answer : (4)

Solution:

Ciliated columnar epithelium consists of columnar cells which have cilia on their free surface. This epithelium is present in the inner surface of hollow organs like fallopian tube (oviducts) and most of the respiratory tract.

(47) Answer : (2)

Solution:

Epithelial tissue

– Has little intercellular matrix

– Does not cover all parts of the body

(48) Answer : (2)

Solution:

Simple squamous epithelium consists of a single thin layer of flattened cells.

(49) Answer : (2)

Solution:

Compound epithelium provides protection against chemical and mechanical stresses.

(50) Answer : (3)

Solution:

Blood flows in blood vessels.

Blood vessels are lined by simple squamous epithelium.

(51) Answer : (4)

Solution:

Bronchioles and fallopian tubes are internally lined by ciliated epithelium. Pancreatic ducts are lined by compound epithelium.

- (52) **Answer :** (2)
Solution:
Epithelium forms covering over all the body organs.
- (53) **Answer :** (4)
Solution:
Cuboidal and columnar cells get specialised for secretion and form glandular epithelium.
- (54) **Answer :** (3)
Solution:
Specialised connective tissues include cartilage, bone, adipose and blood.
- (55) **Answer :** (4)
Solution:
In all connective tissues except blood, the cells secrete fibres of structural proteins called collagen or elastin.
- (56) **Answer :** (3)
Solution:
Cuboidal epithelium is generally found in the tubular parts of nephrons in kidney, duct of glands, thyroid follicles, ovaries and testes.
- (57) **Answer :** (1)
Solution:
Cuboidal epithelium constitutes cube-like cells whereas columnar epithelium is made up of tall cells.
- (58) **Answer :** (2)
Hint:
Identify a loose connective tissue.
Solution:
Adipose tissue stores fats and is located mainly beneath the skin and also around the heart, kidney, eyeballs, etc. Areolar tissue serves as support framework for epithelium and joins skin to muscles and fills the spaces outside organs. Tendons connect muscles to bones and ligaments connect bone to bone.
- (59) **Answer :** (1)
Solution:
Connective tissue is mainly responsible for linking and supporting various tissues and organs of the body.
- (60) **Answer :** (4)
Solution:
Goblet cells are isolated glandular cells of epithelium.